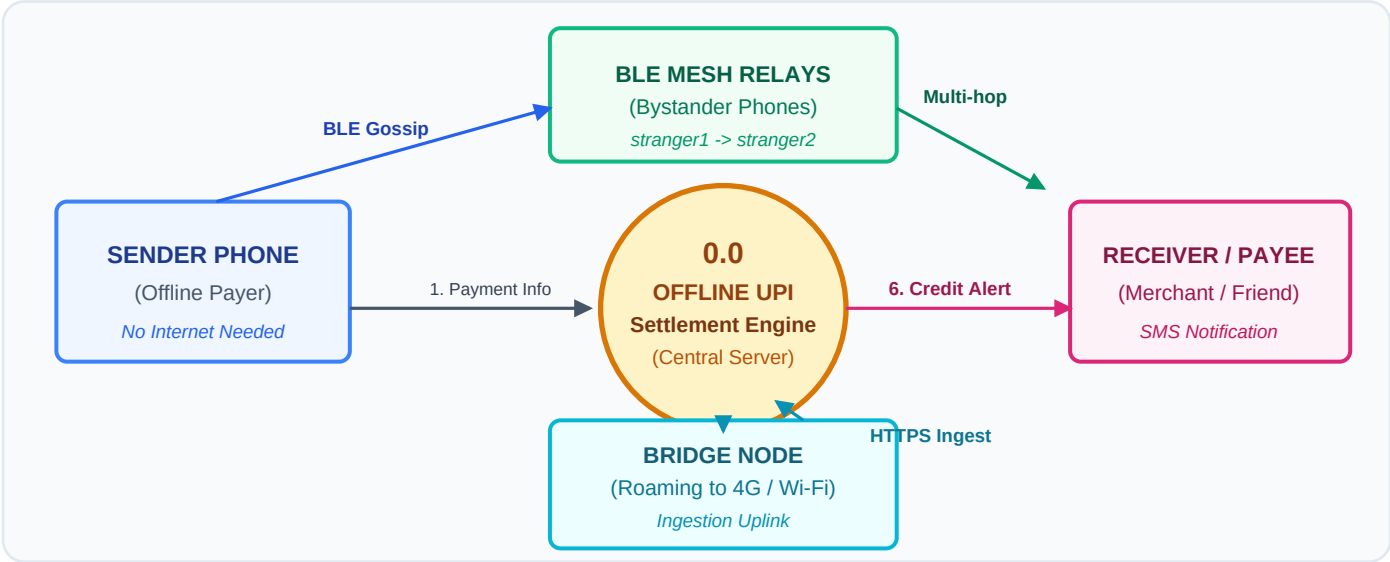


Offline UPI Payment & Settlement System

System Architecture, Data Flow Diagrams (Level 0, 1, 2) and Entity-Relationship (ER) Specifications

1. DFD Level 0 — Context Diagram

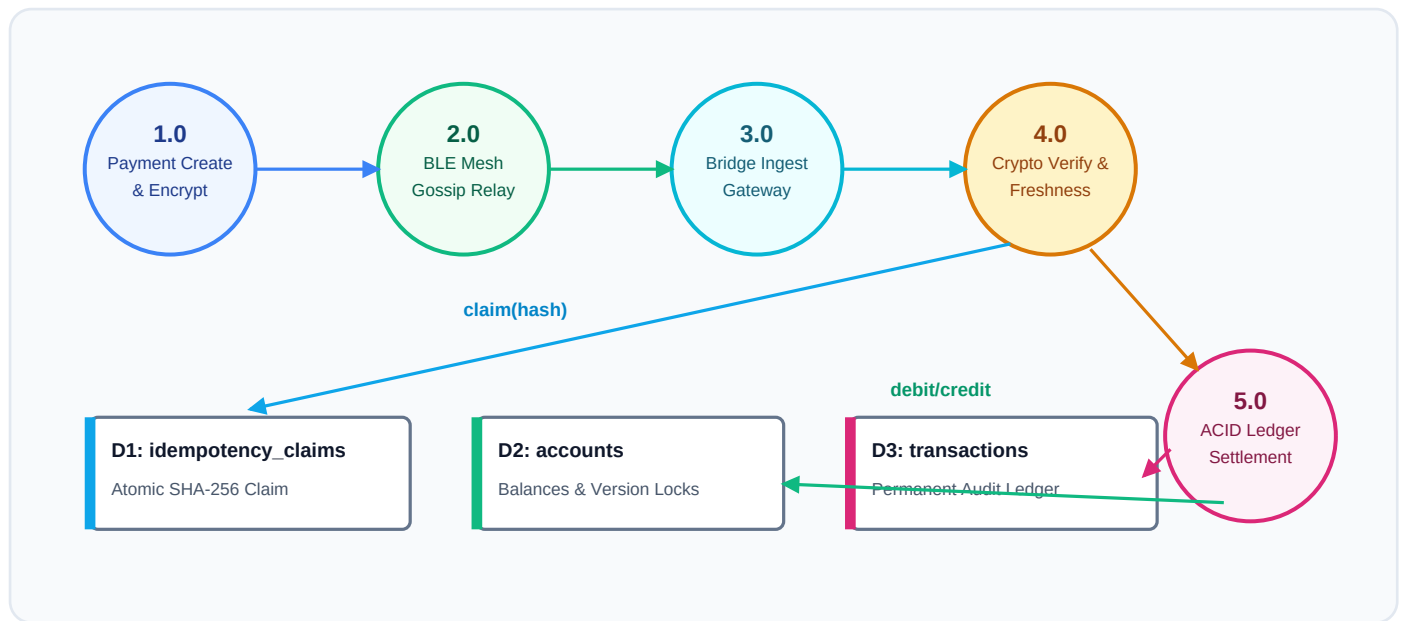
The Level 0 Context Diagram depicts the primary system boundary, showing external entities interacting with the central offline UPI engine without exposing internal storage mechanisms.



Flow ID	Source Entity	Destination	Data Carried	Security / Mechanism
1	Sender Phone	0.0 Client Module	VPA, Amount, PIN, Nonce	Entered 100% offline
2	0.0 Client Module	Sender Storage	Encrypted MeshPacket	RSA-2048-OAEP + AES-GCM
3	Sender Phone	BLE Relays	Ciphertext broadcast	BLE Advertising / GATT
4	BLE Relays	Bridge Node	Relayed MeshPacket	Store-and-forward (hop count)
5	Bridge Node	0.0 Central Server	Ciphertext + Bridge ID	HTTPS POST /api/bridge/ingest
6	0.0 Central Server	Receiver / Payee	Credit SMS / Push Alert	Telecom SMS / Bank Push
7	0.0 Central Server	Bridge Node	Status (SETTLED/DUP)	HTTP 200 / 400 response

2. DFD Level 1 — Subsystem Data Flow Decomposition

Decomposes the system into 5 modular processes across client cryptography, distributed mesh gossip routing, bridge ingestion, and central bank ACID ledger updates.



Processes	Process Name	Input Data	Output Data	Primary Function
1.0	Payment Creation & Encrypt	User input (Amt, PIN)	Signed MeshPacket	Hybrid encryption with Server's RSA Public Key.
2.0	BLE Gossip Routing	MeshPacket (TTL=5)	Relayed packet (TTL--)	Store-and-forward hopping across bystander devices.
3.0	Bridge Ingest Gateway	Cached mesh packets	HTTPS Ingest Request	Uploads ciphertext when device connects to 4G/Wi-Fi.
4.0	Crypto Verification & Deduplication	Ingest payload	Decrypted instruction	Enforces atomic SHA-256 claim, RSA decrypt, GCM tag, and freshness.
5.0	ACID Ledger Settlement	Valid instruction	Ledger record & SMS	Executes atomic debit, credit, optimistic lock and logs audit trail.

3. DFD Level 2 — Ingestion & Settlement Pipeline

Details the exact atomic execution path inside /api/bridge/ingest on the central server:

- **Step 4.1 (SHA-256 Digest):** Central engine hashes incoming ciphertext to produce a 64-char hex digest.
- **Step 4.2 (Atomic Deduplication Gate):** Executes atomic INSERT INTO idempotency_claims. If hash exists, returns DUPLICATE_DROPPED immediately without expensive cryptographic compute.
- **Step 4.3 (Hybrid Decryption & Authentication):** Server unwraps AES key using RSA-2048-OAEP Private Key. Decrypts with AES-256-GCM. 128-bit authentication tag guarantees 0-bit tampering.
- **Step 4.4 (Sliding Freshness Window):** Rejects payments with signedAt older than 24 hours (prevents long-term replay).
- **Step 5.1 & 5.2 (Optimistic Concurrency Balance Updates):** Atomically updates sender balance with WHERE vpa=? AND version=?.
- **Step 5.3 (Ledger Commit):** Appends immutable audit entry to transactions.

4. Entity-Relationship (ER) Diagram & Schema

Relational model enforced in SQLite (upi.db) with primary keys, foreign keys, and version locks.

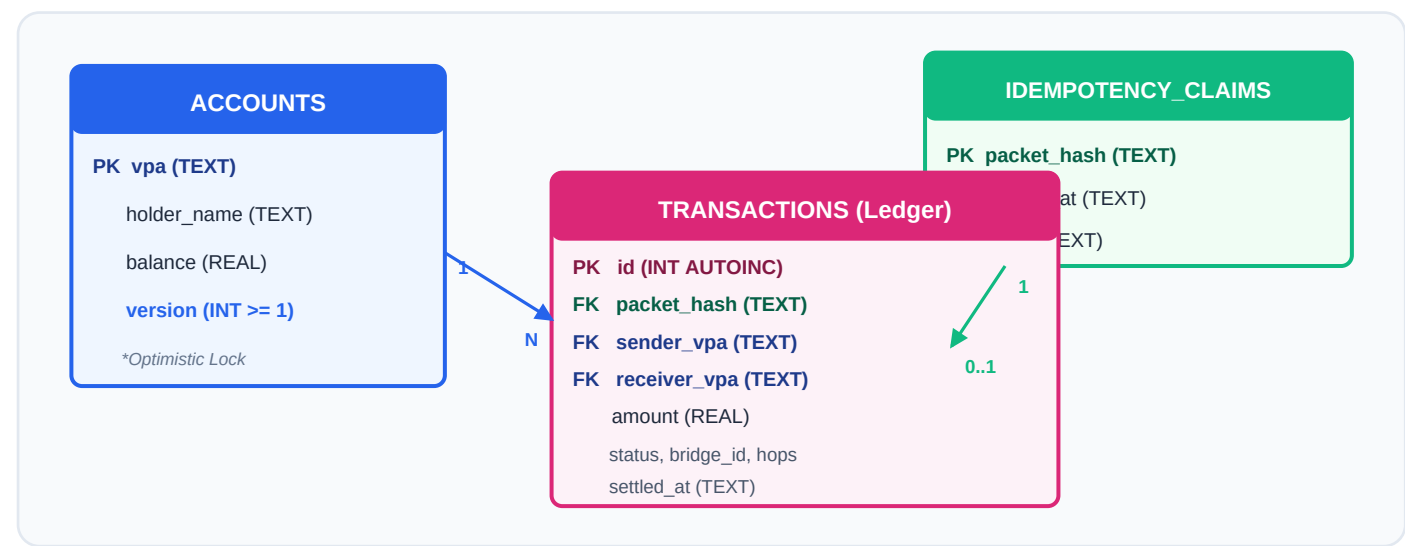


Table Name	Primary Key	Foreign Keys	Attributes & Concurrency Role
accounts	vpa (TEXT)	None	holder_name, balance (REAL), version (INT). Version provides optimistic locking against balance races.
idempotency_claims	packet_hash (TEXT)	None	claimed_at (TEXT), status (TEXT). Acts as atomic putIfAbsent deduplication gate.
transactions	id (INTEGER AUTOINC)	packet_hash, sender_vpa, receiver_vpa	amount (REAL), status, bridge_id, hop_count, settled_at. Permanent audit trail.